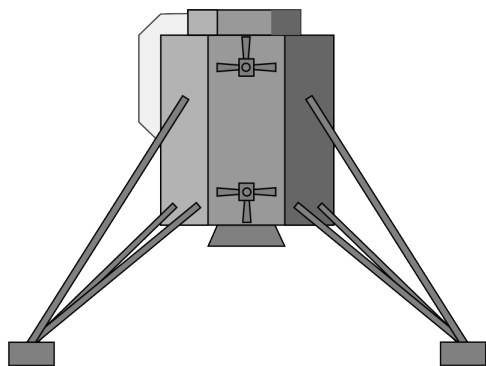




Sky Dart

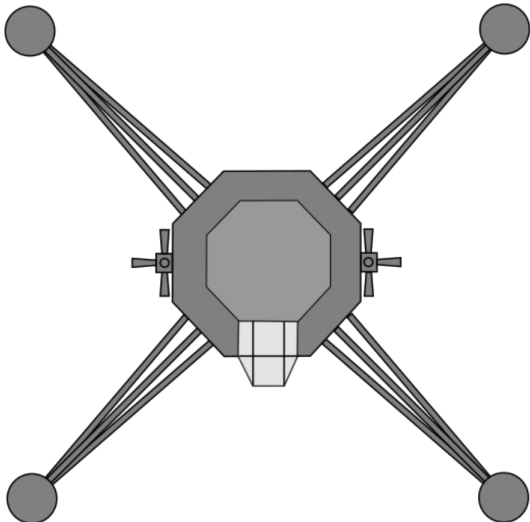


**AMC SKY DART
MODEL: SDART-03**

OPERATORS MANUAL

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SYSTEM INF. (SYS)	- OVERVIEW (OVR) PERFORMANCE (PERF)
PILOT CONTROLS (CTRL)	- HOVER AUTOPILOT (AP HOVR)
SIMULATION (SIM)	- OVERVIEW (OVR) KEYBOARD (KBD) VIRTUAL REALITY (VR) XBOX (XBOX) PLAYSTATION (PS) STEAMDECK (STEAM)



AMC SKY DART

LIGHT-DUTY, SINGLE-SEAT,
SHORT DURATION LOGISTICS
SPACECRAFT FOR CIS-LUNAR
OPERATIONS.

MODEL NUMBER SDART-03

SPECIFICATIONS

EMPTY MASS 2500 KG
MAX TAKEOFF WT 6250 KG
FUEL/OXIDIZER 2500 KG
 (NOMINAL)
FUEL RESERVE 1100 KG
 (3600 KG MAX)

PROPULSION

ENGINE MAX THR 20 KN
ENGINE MIN THR 2 KN

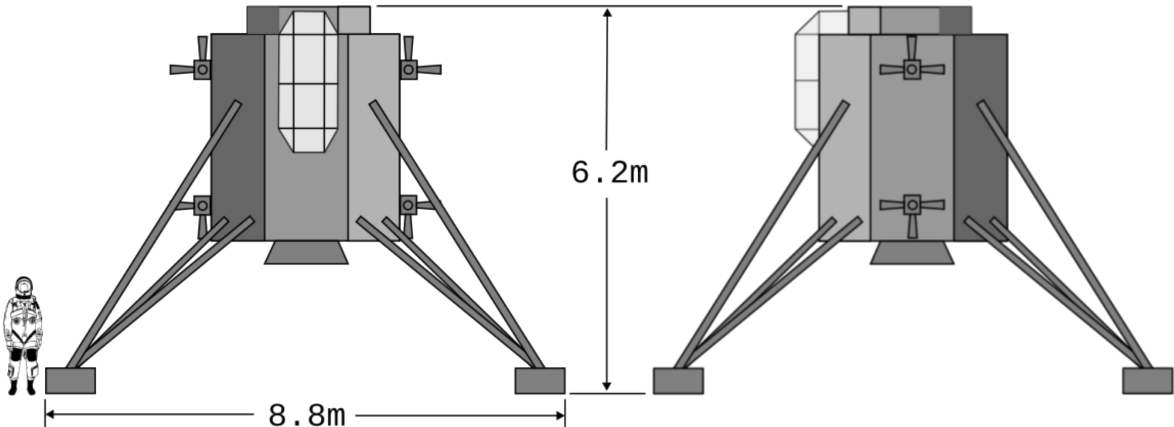
DIMENSIONS

WIDTH OVERALL 8.8 M
HEIGHT OVERALL 6.2 M

LANDING GEAR

MAX LANDING SPD (EMPTY) 4.7 M/S
MAX LANDING SPD (MTOW) 2.3 M/S

NOTE: MAX LANDING SPEEDS ASSUME
SIMULTANEOUS TOUCHDOWN OF ALL
FEET AND ZERO LATERAL OR
ROTATIONAL VELOCITY.





SYS
SYSTEM INF.

PERF
PERFORMANCE

SKY DART PERFORMANCE DATA
SAFE MASS ENVELOPE 01/11

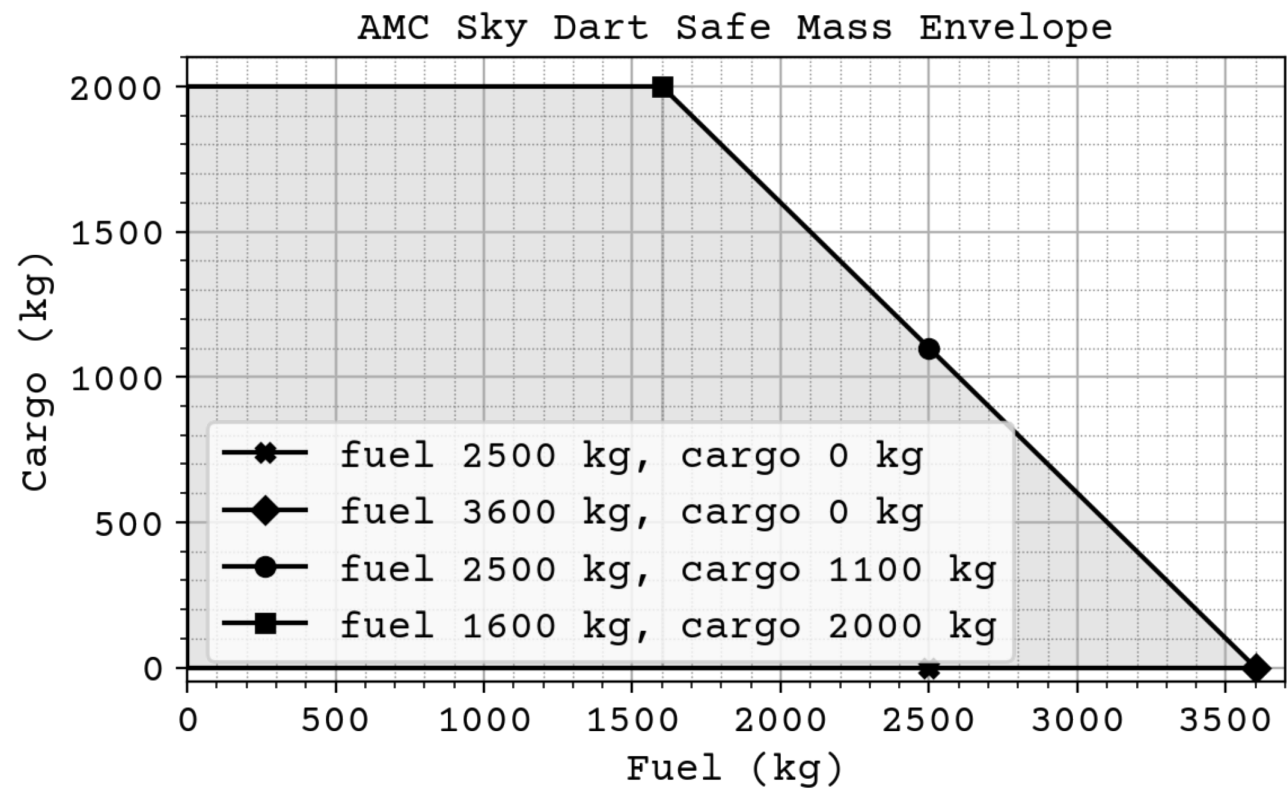
NOTES ON MASS ENVELOPE

ORE CARGO IS HARD-LIMITED
TO 2000 KG

ABSOLUTE MAXIMUM FUEL
CAPACITY IS 3600 KG

NOMINAL MAXIMUM FUEL
CAPACITY IS 2500 KG

OPERATION OUTSIDE THE
SHADED REGION IS PHYSICALLY
POSSIBLE BUT STRONGLY
DISCOURAGED. RISKS OF
LANDING GEAR DAMAGE OR
AUTOPILOT CONTROL FAILURE
ARE SUBSTANTIAL AND MAY
RESULT IN LOSS OF
SPACECRAFT, PILOT DEATH,
AND EVEN INCUR A
**SIGNIFICANT RISK OF VOIDING
THE MANUFACTURER WARRANTY.**



WARN

RATE
MODE

HOVR
MODE

xx.x
xxxx

VS
ALT

HSPD
MODE

PROP
CTRL

XPTR
AUTO

ALT
CTRL

AUTOPILOT WARNING INDICATOR

Red light illuminates and siren warns the pilot of EXTEME DANGER due to one of following:

- 1. Hull Breach or Loss of Atmosphere
- 2. Landing Gear Failure (See: DIAG/L-GEAR in FMS for details)
- 3. Uncommanded Autopilot Disconnect.

In the event of WARN activation you may silence the alarm/light by pressing the button then IMMEDIATELY ascertain the source of the problem.

AUTOPILOT ROTATIONAL MODE

CONTROL	STATE	SUMMARY OF FUNCTION
-----	-----	-----
RATE MODE	ACTIVE	Pilot joystick commands pitch/roll rate (see ALT CTRL).
	INACTIVE	Pilot joystick commands pitch/roll torque.

NOTES

- 1. RATE MODE is automatically ACTIVATED upon lifting off from the surface and DEACTIVATED when landing.
- 2. With RATE MODE active, centering the pilot joystick will stop all rotational motion.
- 3. RATE MODE MUST be ACTIVE before activating HOVR MODE.

WARN

RATE
MODE

HOVR
MODE

xx.x

xxxx

VS
ALT

HSPD
MODE

PROP
CTRL

XPTR
AUTO

ALT
CTRL

AUTOPILOT VERTICAL MODE

CONTROL	STATE	SUMMARY OF FUNCTION
-----	-----	-----
HOVR MODE	ACTIVE	Autopilot maintains commanded vertical rate. Pilot controls the rate using vernier knob. Move vernier forward to increase positive rate, backward to increase negative rate. Autopilot adjusts the main engine thrust to achieve desired rate. Moving the throttle t-stick with HOVR MODE active will cause an Uncommanded Autopilot Disconnect WARNING.
	INACTIVE	Pilot must use the MANUAL throttle t-stick control and/or the vernier knob to control main engine thrust while monitoring the radar vertical rate display.
DISPLAY		

VS		Displays commanded vertical rate in meters/sec. With HOVR MODE inactive, commanded vertical rate displays inactive/all dashes.
ALT		Displays commanded altitude in meters when VS=0. With a non-zero VS value set, commanded altitude displays inactive/all dashes.

NOTE: RATE MODE MUST be ACTIVE before activating HOVR MODE.

WARN

RATE
MODE

HOVR
MODE

xx.x

xxxx

VS
ALT

HSPD
MODE

PROP
CTRL

XPTR
AUTO

ALT
CTRL

AUTOPILOT HORIZONTAL MODE

CONTROL	STATE	SUMMARY OF FUNCTION
-----	-----	-----
HSPD MODE	ACTIVE	Autopilot maintains commanded horizontal rates. Pilot controls the rates using the joystick. The commanded rates are indicated by the green diamond bugs on the crosspointer display. Range buttons on the crosspointer X5 and X50 correspond to ±5 m/s and ±50 m/s.
	INACTIVE	Pilot must use the joystick to MANUALLY control the ship attitude while monitoring the radar horizontal rates and crosspointer needles.
PROP CTRL	ACTIVE	With HSPD MODE active, the green diamond bugs on the crosspointer display will match the joystick position. NOTE: Releasing or centering the joystick with PROP CTRL active will make the autopilot drive the horizontal speed to zero.
XPTR AUTO	ACTIVE	Radar crosspointer range is chosen automatically based on drift rate or drift command (green diamonds) (with HSPD MODE active).
	INACTIVE	Radar crosspointer range must be selected manually.

WARN

RATE
MODE

HOVR
MODE

xx.x
xxxx

VS
ALT

HSPD
MODE

PROP
CTRL

XPTR
AUTO

ALT
CTRL

AUTOPILOT ALTERNATE CONTROL MODE

CONTROL	STATE	SUMMARY OF FUNCTION
ALT CTRL	ACTIVE	Joystick left/right motion controls YAW rotation only.
	INACTIVE	HSPD INACTIVE: Joystick left/right motion controls ROLL rate. HSPD ACTIVE: Joystick left/right motion controls lateral rate.

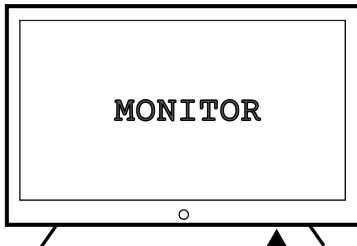
AUTOPILOT RADAR SPECIFICATIONS OVERVIEW:

RADAR	DIRECTION	MAX RANGE	MAX DOPPLER	NOTES
OMNI	DOWN (1)	400 m	200 m/s	Altitude and vertical velocity
DIRECTIONAL	DOWN (1)	2000 m	N/A	Altitude only (low fidelity)
DIRECTIONAL	SIDE (4)	2000 m	200 m/s	Effective lateral range 500 m

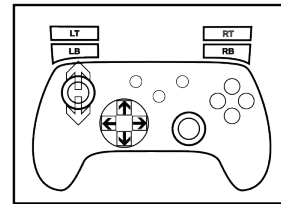
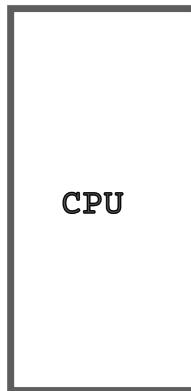
AUTOPILOT LIMITATIONS

1. HOVR MODE is available between zero and 90% of the OMNI MAX RANGE or 360 m altitude.
2. Maximum pitch/roll in HSPD MODE is 20 degrees, derated to 5 degrees at 360 m altitude.
3. HOVR MODE will automatically limit vertical rate to stay in 0-360m altitude range.
4. Excursion outside of the autopilot operational limits will cause an automatic Uncommanded Autopilot Disconnect and activate the AUTOPILOT WARNING INDICATOR.

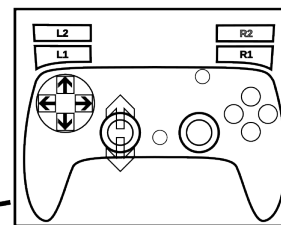
**COLOR HI
RES
DISPLAY**



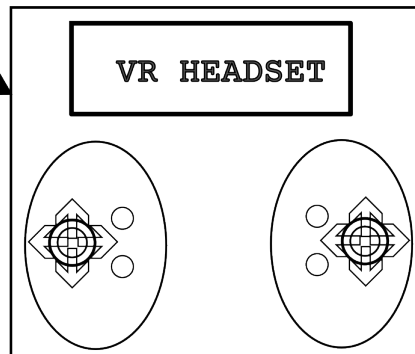
**INTEL OR AMD BASED
WORK-STATION WITH
LINUX OR WINDOWS
10/11 OPERATING
SYSTEM AND
DISCRETE GPU SUB-
SYSTEM**



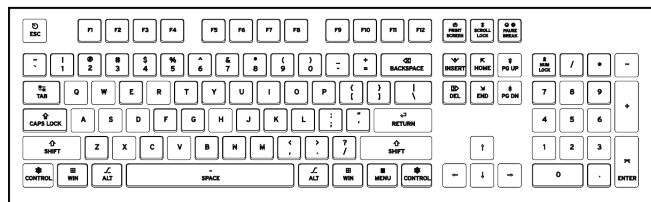
**XBOX CONTROL
SEE: SIM XBOX**



**PS CONTROL
SEE: SIM PS**

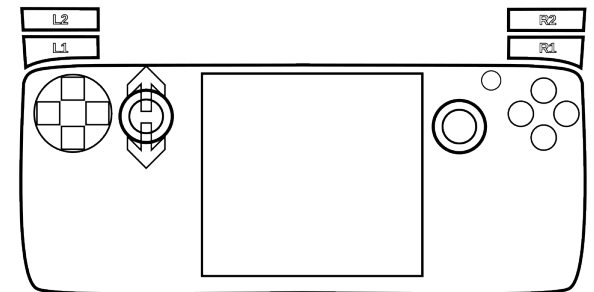


SEE: SIM VR



SEE: SIM KBD

**ALTERNATIVE HAND-HELD
SIMULATION CPU: STEAM-DECK
MANUF. BY VALVE**



SEE: SIM STEAM



KEYBOARD CONTROLS

- MOVEMENT:**
 - W: PAN UP
 - S: PAN DOWN
 - A: PAN (MOVE L)
 - D: PAN (MOVE R)
 - V: RESET PILOT VIEW
- CAMERA:**
 - C: TOGGLE CAMERA VIEWS
 - 1: INTERIOR COCKPIT
 - 2: EXTERIOR LOCK TO SHIP
 - 3: EXTERIOR LOCK TO HORIZON
 - 4: EXTERIOR LOCK TO GROUND
- GAME MENU:**
 - ESC: GAME MENU
- COCKPIT:**
 - X or KP 5: TOGGLE ATTITUDE HOLD
 - H: TOGGLE HOVER
 - L: TOGGLE LANDING LIGHT
 - SHIFT: TOGGLE ALTERNATE CONTROLS
- JOYSTICK:**
 - KP 1: YAW LEFT
 - KP 2: PITCH BACKWARD
 - KP 3: YAW RIGHT
 - KP 4: ROLL LEFT
 - KP 5: ROLL RIGHT
 - KP 6: PITCH FORWARD
 - KP 7: ROLL LEFT + YAW LEFT
 - KP 8: ROLL RIGHT + YAW RIGHT
- MAIN ENGINE:**
 - PG UP: MAXIMUM THRUST
 - PG DN: MINIMUM THRUST
 - PG UP +: THRUST VERNIER +
 - PG DN -: THRUST VERNIER -

DEFAULT MOUSE CONTROLS

- USE MOUSE LEFT-CLICK TO PRESS BUTTONS OR GRAB THINGS.
- HOLD MOUSE RIGHT-CLICK AND MOVE TO PAN COCKPIT VIEW.

1. USE MOUSE LEFT-CLICK TO PRESS BUTTONS OR GRAB THINGS.
2. HOLD MOUSE RIGHT-CLICK AND MOVE TO PAN COCKPIT VIEW.
3. HOLD MOUSE RIGHT-CLICK AND KEYBOARD WASD TO MOVE COCKPIT VIEW.
4. POSITION MOUSE POINTER OVER THROTTLE CONTROL AND USE SCROLL WHEEL TO ADJUST THE THROTTLE SETTING.

GENERIC VR CONTROLLER MAP

MAIN ENGINE

THRUST
VERNIER +

YAW

THRUST
VERNIER -

MENU

RESET
PILOT VIEW

LEFT HAND

RIGHT HAND

ROLL
LEFT

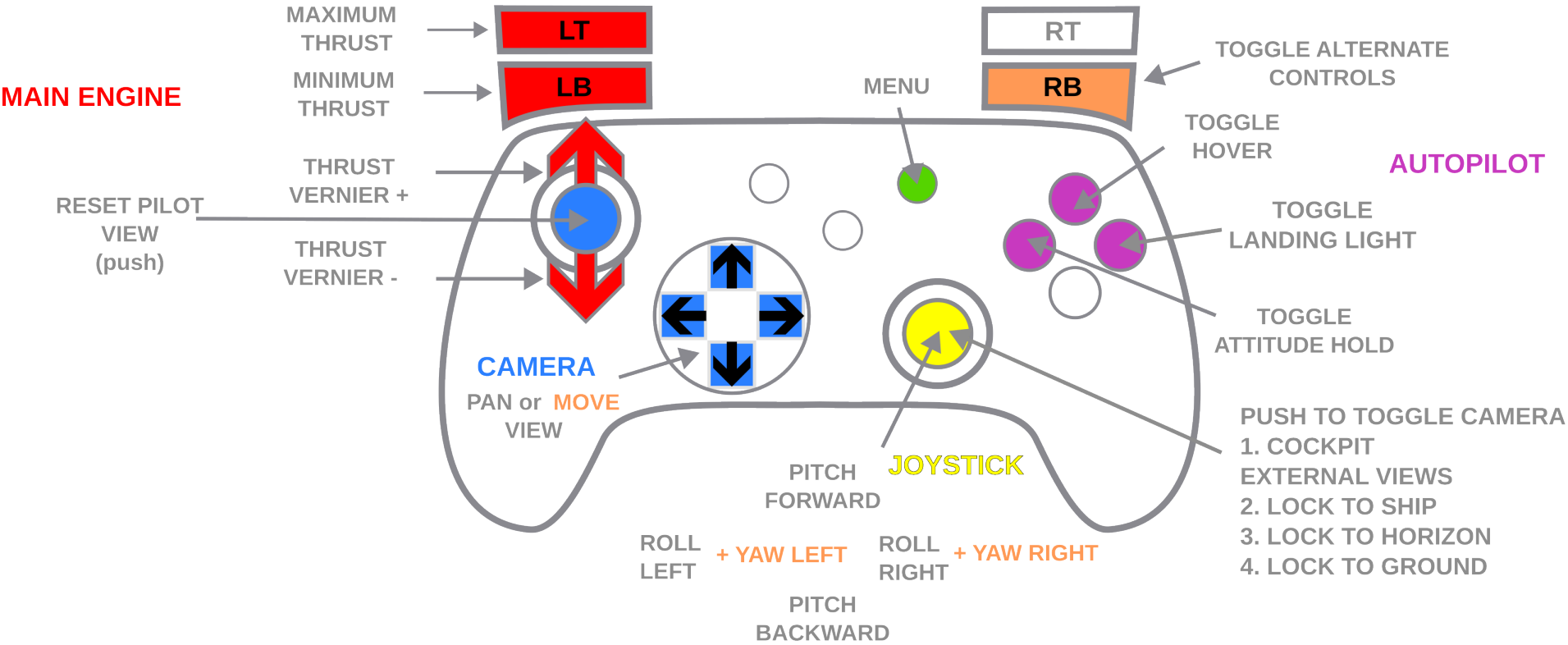
ROLL
RIGHT

PITCH
FORWARD

PITCH
BACKWARD

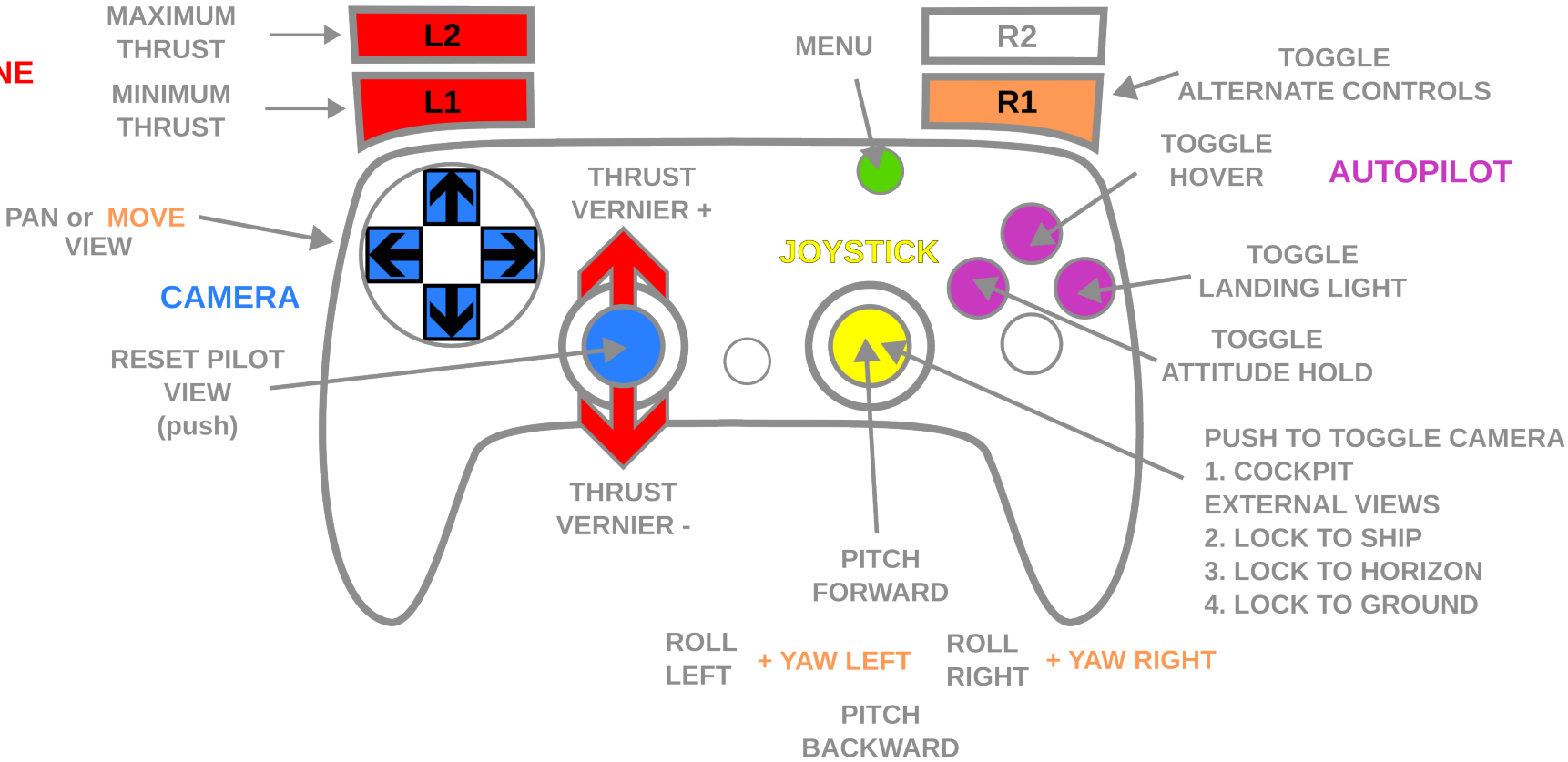
JOYSTICK

XBOX CONTROLLER MAP



PLAYSTATION CONTROLLER MAP

MAIN ENGINE



STEAM DECK CONTROLLER MAP

